

Data-Driven Upwork Portfolio Planning

Analysis of Freelance Data Jobs and Portfolio Project Design

Lénárt Martin

Portfolio Project Documentation

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1 Introduction

1.1 Background

Digital labour platforms have significantly transformed the global labour market over the past decade by enabling organisations and independent professionals to collaborate across geographical boundaries. These platforms provide businesses with flexible access to specialised expertise while simultaneously creating new employment opportunities for freelancers worldwide. Among the different forms of platform-mediated work, online freelancing platforms have become a particularly important segment of the digital labour market by connecting businesses with specialised professionals worldwide [1], [2].

Online freelancing platforms play a central role within the digital labour market by enabling knowledge-intensive services such as software development, data analysis, graphic design and consulting to be delivered entirely online. The OECD identifies platforms such as Upwork, Freelancer and PeoplePerHour as representative examples of online freelancing platforms [1].

Upwork is one of the most prominent online freelancing platforms, connecting businesses with independent professionals across a broad range of knowledge-intensive disciplines. Its marketplace covers fields including software development, data science, engineering, design, writing and consulting, making it representative of the highly skilled segment of digital platform work. Owing to its broad occupational coverage and global marketplace, Upwork provides a relevant case for examining labour market trends, skill demand project characteristics within the online freelancing economy [1], [3].

1.2 Motivation

Although online freelancing platforms provide access to a global marketplace, entering these platforms can be challenging for newcomers. Beyond possessing technical expertise, new freelancers must establish credibility in an environment where client decisions are strongly influenced by previous work history, portfolio quality and platform reputation. At the same time, rapidly evolving technologies and changing client requirements require continuous adaptation, making it difficult for beginners to determine which skills are most valuable to develop and how they should position themselves within the market [4]–[6].

Consequently, aspiring freelancers are faced with several practical questions before

entering the market. Which technical skills are currently in demand? What types of portfolio projects best demonstrate professional competence? Which market segments offer realistic opportunities for beginners? How can professional credibility be established without an existing client history? Addressing these questions requires a systematic understanding of current market conditions rather than relying solely on personal assumptions or anecdotal advice [4], [7].

The present project was motivated by the need to address these challenges through a data-driven analysis of the Upwork marketplace. Rather than relying on subjective opinions or informal recommendations, the project investigates publicly available job marketplace to identify market trends, demanded skills and portfolio opportunities that may support prospective freelancers in making more informed decisions about their professional development.

1.3 Project Objectives

The primary objective of this project is to provide a data-driven analysis of the current Upwork marketplace within selected data-related freelance categories by examining publicly available marketplace data. The study focuses on areas including data analysis, data engineering, Python automation and machine learning in order to provide an objective overview of current market demand, required skills and project characteristics within this segment of the online freelancing market.

To achieve this objective, the project involves the systematic collection, organisation and analysis of publicly available marketplace data from the selected Upwork categories. Particular attention is given to identifying the most frequently requested technical skills, analysing job budgets and pricing models, examining the distribution of job types and exploring relationships between different marketplace characteristics. The results are presented using descriptive statistical methods and clear visualisations in order to provide an accessible overview of the analysed market segment. The project is intended to provide a descriptive analysis of the current marketplace rather than to predict future labour market trends.

The expected outcome of the project is a comprehensive overview of the selected segment of the Upwork marketplace based on objective, data-driven analysis. By identifying current demand for technical skills, common project characteristics and prevailing pricing patterns, the study aims to provide practical insights that may support aspiring freelancers in making more informed decisions about skill development, portfolio design and market positioning. Furthermore, the project demonstrates a structured workflow for collecting, processing and analysing online marketplace data, providing a methodological framework that can be adapted to similar marketplace analyses.

1.4 Research Questions

To achieve the objectives described in the previous section, the project addresses the following research questions concerning the selected Python-related data freelance categories on the Upwork marketplace:

1. Which types of projects are most frequently posted?
2. Which Python libraries and frameworks are most commonly required?
3. Which technologies and software tools are most frequently required?
4. Which combinations of technologies are most frequently requested together?
5. What budget ranges and hourly rates are associated with the analysed freelance jobs?
6. Which portfolio projects would best reflect current client demand?

Each research question is addressed in a dedicated section of the exploratory data analysis, while the final question synthesises the findings into a practical portfolio development strategy.

1.5 Document Structure

The remainder of this report is organised into four main phases that reflect the overall workflow of the project. Following this introduction, Chapter 2 presents the data collection process, including the definition of the analysed market segment, the data acquisition procedure and the preparation of the dataset for subsequent analysis.

Chapter 3 contains the exploratory data analysis and is organised around the research questions introduced in the previous section. Each research question is addressed individually through descriptive statistical analyses and data visualisations in order to provide a comprehensive understanding of the selected Upwork market segment.

Building upon these findings, Chapter 4 synthesises the analytical results into broader market insights, identifies the most valuable technical skills and technologies, and proposes practical recommendations for aspiring freelancers. Chapter 5 translates these findings into a portfolio development strategy by recommending a sequence of portfolio projects designed to maximise their relevance to current market demand.

Finally, Chapter 6 summarises the main findings of the project, discusses its limitations and outlines potential directions for future work.

2 Phase 1 - Data Collection

2.1 Objectives

The objective of the first phase is to establish a reliable and reproducible dataset that serves as the empirical foundation for the subsequent stages of the project. Since the validity of any data-driven analysis depends fundamentally on the quality and consistency of the underlying data, this phase focuses on collecting, organising and preparing relevant information from the selected Upwork marketplace categories.

The outcome of this phase is a structured dataset containing publicly available job information related to selected Python-related data freelance categories, including data analysis, data engineering, Python automation and machine learning. The resulting dataset is prepared for exploratory data analysis and provides the basis for answering the research questions introduced in Chapter 1.

2.2 Scope of the Dataset

The scope of the dataset was deliberately limited to a selected segment of the Upwork marketplace consisting of Python-related data freelance jobs. Rather than analysing the entire marketplace, the study focuses on categories including data analysis, data engineering, Python automation and machine learning. This targeted scope allows the analysis to concentrate on a coherent group of closely related professional domains while remaining directly relevant to the objectives of the project.

The final dataset contains 44 manually collected Upwork job postings gathered during July 2026. Each job was stored as an individual text file containing the complete textual content of the corresponding job posting together with additional metadata, including the original job URL, the selected search category and the search keyword used during data collection. Preserving the complete job descriptions ensured that all potentially relevant information remained available throughout the subsequent stages of preprocessing and analysis.

The relatively small size of the dataset reflects the current limitations of publicly accessible Upwork data rather than a limitation of the developed data collection workflow. Since direct programmatic access to the Upwork platform was not available, each job posting had to be collected manually. Nevertheless, the developed data collection pipeline was intentionally designed in a modular manner. With only minor modifications to its initial acquisition stage, the same workflow could be applied to substantially larger datasets if direct access to the Upwork API or another

automated data source became available in the future.

2.3 Data Collection Procedure

The data collection process began by identifying relevant job postings within the selected Upwork categories. Jobs related to data analysis, data engineering, Python automation and machine learning were located using the Upwork search interface and predefined search terms corresponding to the scope of the study. Each selected job was then individually reviewed to determine whether it matched the objectives of the dataset.

Since direct programmatic access to the Upwork platform was not available, each job was collected manually. The complete textual content of every selected job posting was copied into an individual plain text file, preserving the original structure of the advertisement. In addition to the job description, each file contained supplementary metadata such as the original job URL, the selected marketplace category and the search keyword used during data collection.

Rather than extracting only predefined variables during data collection, the complete job descriptions were preserved. This approach ensured that no potentially relevant information was discarded at an early stage and allowed the preprocessing pipeline to identify additional technologies, software tools, programming libraries and technical skills directly from the original job descriptions.

The adopted collection procedure separates raw data acquisition from subsequent preprocessing and analysis. As a result, the original job postings remain permanently available for verification, while improvements to the preprocessing pipeline can be applied repeatedly without requiring the data collection process to be repeated. This modular workflow improves the reproducibility and maintainability of the entire project.

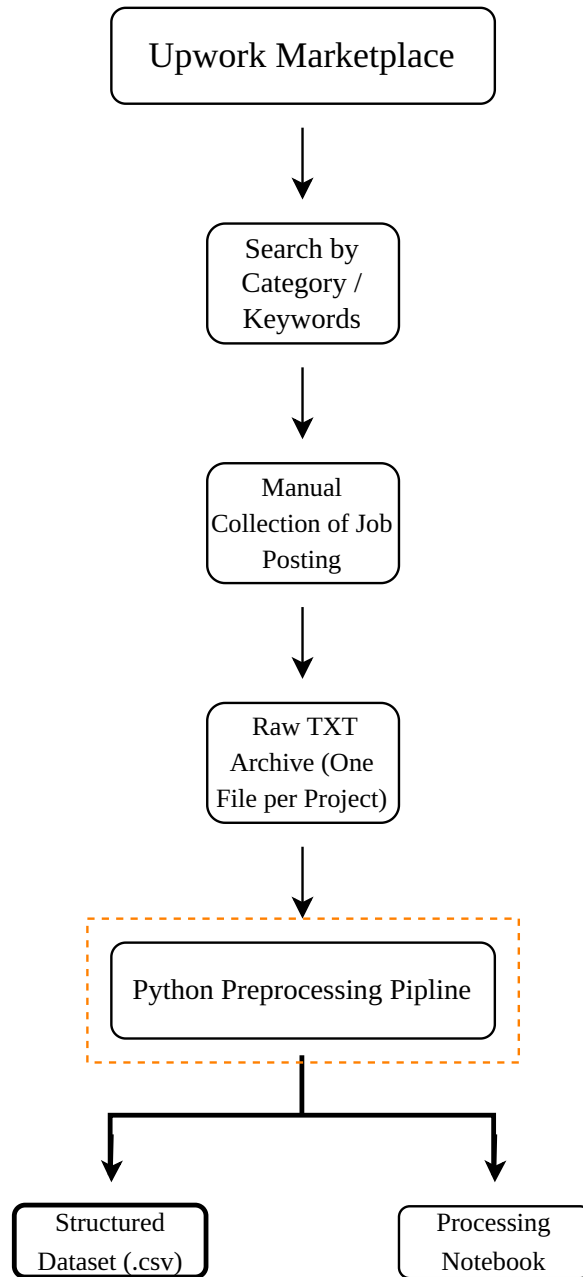


Figure 2.1: Overview of the data collection workflow.

2.4 Dataset Structure

Following the data collection procedure, the raw text archive was transformed into a structured dataset suitable for automated analysis. During this process, relevant information was extracted from each job posting and organised into a tabular format, where each row represents a single Upwork job and each column corresponds to a specific job attribute. The resulting structure enables efficient filtering, statistical analysis and visualisation while preserving the information required to answer the research questions introduced in Chapter 1.

Table 2.1: Structure of the final dataset.

Variable	Type	Description
job_id	Integer	Unique identifier assigned to each collected job.
category	Categorical	Upwork marketplace category from which the job was collected.
search_keyword	String	Search keyword used during data collection, if applicable.
title	String	Title of the job posting.
description	Text	Complete textual description of the project.
mandatory_skills	Text	Technical skills explicitly required by the client.
preferred_qualifications	Text	Skills or qualifications listed as desirable but not mandatory.
experience_level	Categorical	Client’s requested experience level.
job_type	Categorical	Hourly or fixed-price job.
project_type	Categorical	Type of project (e.g. ongoing project).
duration	String	Expected project duration.
workload	String	Expected weekly workload.
budget_type	Categorical	Representation of the available budget information.
budget_min	Numeric	Lower budget bound or hourly rate.
budget_max	Numeric	Upper budget bound or hourly rate.
posted_date	String	Relative publication date shown on Upwork.
proposals	String	Number of submitted proposals.
job_url	String	Original URL of the Upwork job posting.
collected_at	DateTime	Timestamp indicating when the job was collected.

The dataset combines unstructured textual information with structured metadata extracted from each job posting. While the textual fields preserve the original content required for keyword extraction and qualitative analysis, the structured vari-

ables provide the basis for statistical analyses, filtering and visualisation throughout the subsequent stages of the project.

2.5 Data Cleaning and Preprocessing

Although the manually collected text files contained all information required for the subsequent analysis, the data were stored as unstructured text and therefore required preprocessing before quantitative analysis could be performed. The primary objective of this stage was to transform the raw text archive into a consistent, structured dataset while preserving the original job information.

A Python-based preprocessing pipeline was developed to automate the extraction of relevant job attributes from each text file. Individual extraction functions were implemented for all major project characteristics, including the job title, category, job description, required skills, experience level, project type, budget information, workload, duration, proposal statistics and the original job URL. This modular design improves the maintainability of the preprocessing pipeline while also allowing it to be extended or adapted to future changes in the structure of Upwork job postings. The extracted values were subsequently organised into a unified tabular structure matching the predefined dataset schema.

Several preprocessing steps were incorporated to improve the consistency of the resulting dataset. Budget values were converted into a unified numerical representation, categorical variables were standardised, missing values were handled explicitly where applicable, and the collected information was validated before being exported to the final CSV dataset. Automating these operations ensured that identical processing rules were applied to every collected job, thereby improving the reproducibility, consistency and maintainability of the entire preprocessing workflow.

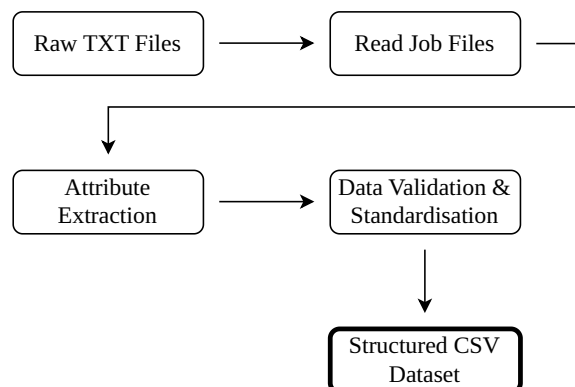


Figure 2.2: Overview of the preprocessing pipeline used to transform raw Upwork job postings into a structured dataset.

2.6 Phase Summary

The first phase of the project established a complete and reproducible workflow for collecting and preparing Upwork job postings for quantitative analysis. Starting from manually collected raw text files, the developed preprocessing pipeline transformed unstructured job descriptions into a structured dataset containing both textual information and standardised job attributes. Throughout this phase, particular emphasis was placed on data consistency, modularity and reproducibility, ensuring that the resulting dataset provides a reliable foundation for subsequent analysis.

With the data collection and preprocessing stages completed, the project proceeds to exploratory data analysis. The following chapter addresses each of the research questions introduced in Chapter 1 through descriptive statistical analyses of the structured dataset, providing the empirical basis for the market insights and portfolio recommendations presented in the subsequent phases.

3 Phase 2 - Exploratory Data Analysis

The structured dataset created during Phase 1 serves as the basis for the exploratory analysis presented in this chapter. The objective of this phase is to answer the research questions introduced in Chapter 1 through descriptive statistical analyses of the collected Upwork job postings. The results presented here describe the observed characteristics of the analysed dataset, while their broader interpretation and practical implications are discussed in the following chapter.

Prior to addressing the individual research questions, each job was assigned to one or more predefined job categories using a rule-based keyword matching approach. Four primary categories were considered throughout the analysis: Data Engineer, Data Analyst, Machine Learning and Python Automation. The classification was performed by matching predefined keywords against the job title, description and original Upwork category. Certain keywords were treated as stronger indicators and therefore contributed a higher classification score, while jobs that did not satisfy the minimum matching criterion were assigned to the category *Other*.

The resulting classifications were stored as an additional attribute of the structured dataset, allowing every job to be associated with one or more job categories. This enriched dataset provides the foundation for all subsequent analyses presented in this chapter, including the investigation of job distributions, required technologies, technical skills and budget characteristics.

3.1 Research Question 1

Question

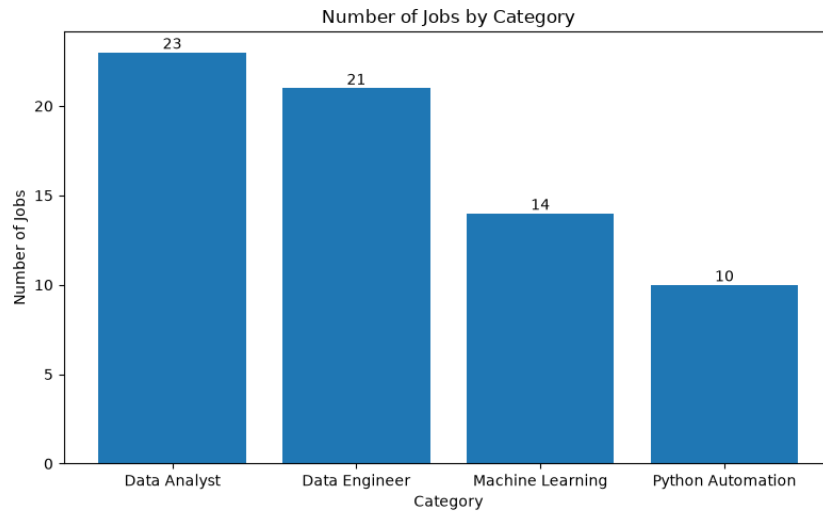
Which types of projects are most frequently posted?

Methodology

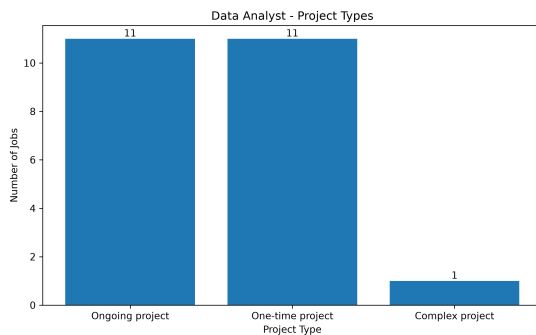
Following the job classification procedure described at the beginning of this chapter, the project type of every classified job posting was extracted from the structured dataset. Three project types were considered throughout the analysis: *Ongoing project*, *One-time project* and *Complex project*. The frequency of each project type was then calculated separately for every job category in order to identify differences in project characteristics across the analysed Python-related freelance domains.

Figure 3.1a presents the number of classified job in each job category, while Fig-

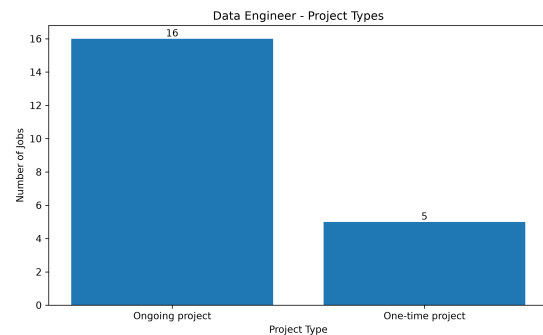
ures 3.1b–3.1e illustrate the distribution of project types within each category.



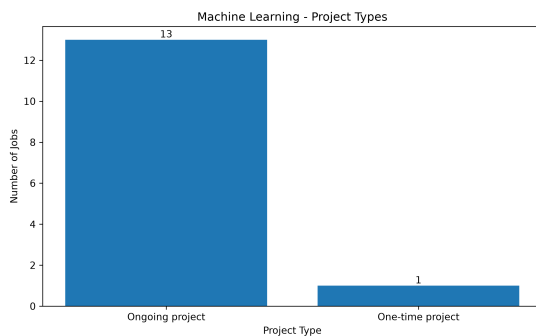
(a) Number of classified jobs by job category.



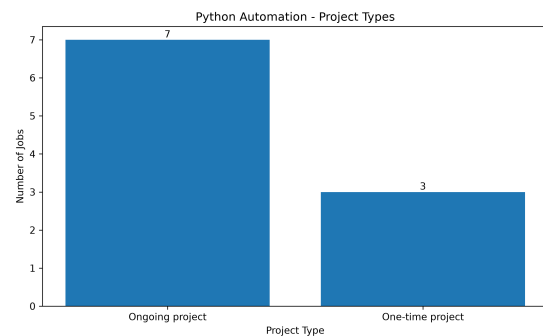
(b) Data Analyst



(c) Data Engineer



(d) Machine Learning



(e) Python Automation

Figure 3.1: Distribution of job categories and project types within the analysed dataset. The complete numerical data used to generate these figures are available in */data/exports/tables/project – type – summary.csv*

Results

The classified dataset contains 23 Data Analyst, 21 Data Engineer, 14 Machine

Learning and 10 Python Automation jobs (Figure 3.1a). Although the analysed dataset is relatively small, these distributions indicate that data-oriented freelance work is primarily concentrated in the Data Analyst and Data Engineer categories.

Project type distributions reveal a clear preference for ongoing collaborations in most categories. Data Engineer jobs exhibit the strongest tendency towards long-term engagements, with 16 ongoing projects compared to only 5 one-time projects. Machine Learning jobs show an even more pronounced pattern, where 13 out of 14 classified jobs correspond to ongoing work. Python Automation jobs follow a similar trend, consisting of 7 ongoing and 3 one-time projects.

Data Analyst jobs differ slightly from the other categories. While ongoing projects remain common, the numbers of ongoing and one-time projects are identical (11 each), and one job was classified as a complex project. This makes Data Analyst the only analysed category containing all three project types.

Discussion

The analysis suggests that long-term collaborations dominate the selected Python-related freelance market. This tendency is particularly pronounced for Data Engineering and Machine Learning jobs, where clients appear to prefer sustained cooperation rather than isolated tasks. In contrast, Data Analyst projects exhibit a more balanced distribution between ongoing and one-time engagements, indicating a broader variety of project scopes within this category. These observations provide an initial overview of the analysed marketplace and establish the context for the subsequent investigation of required technologies, Python libraries and technical skills.

3.2 Research Question 2

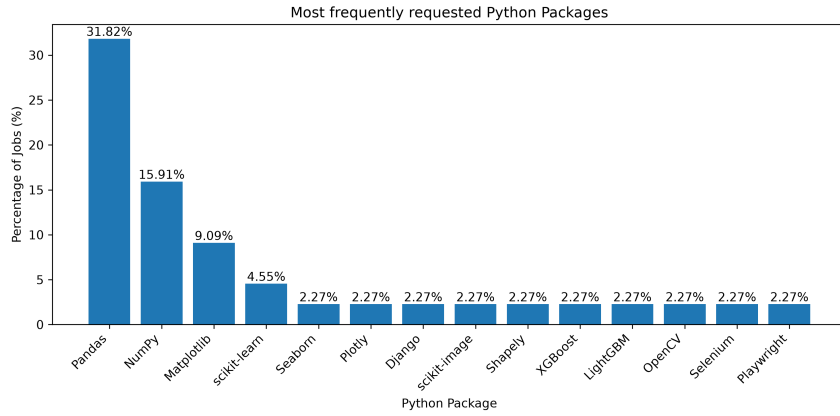
Question

Which Python libraries and frameworks are most commonly required?

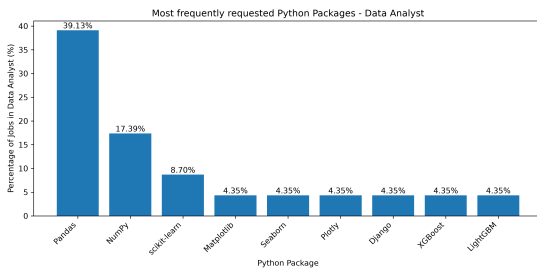
Methodology

The structured dataset was analysed to identify explicit references to Python libraries and frameworks in the job descriptions. Package names were extracted using a predefined dictionary of commonly used Python libraries relevant to data analysis, data engineering, machine learning and automation. Whenever a package was mentioned in a job description, it was counted once for that job, regardless of the number of occurrences within the text.

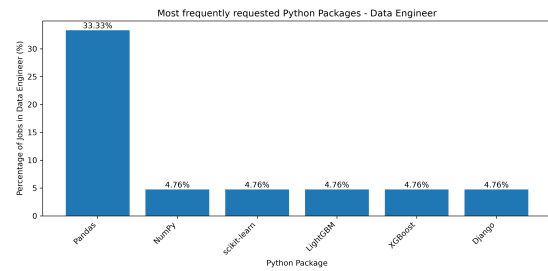
To facilitate comparison between categories of different sizes, package frequencies were normalised by the number of classified jobs in each category and expressed as percentages. Figure 3.2a summarises the overall frequency of Python packages within the analysed dataset, while the accompanying subfigures 3.2b–3.2e present the distributions separately for Data Analyst, Data Engineer, Machine Learning and Python Automation jobs.



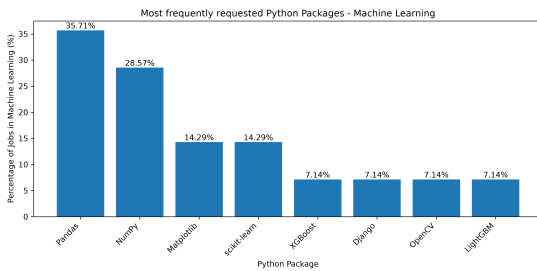
(a) Overall frequency of Python packages.



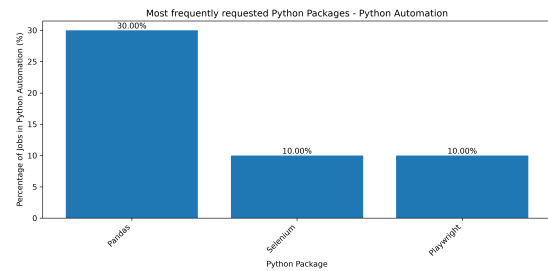
(b) Data Analyst



(c) Data Engineer



(d) Machine Learning



(e) Python Automation

Figure 3.2: Frequency of Python libraries and frameworks within the analysed dataset, overall and by job category. The complete numerical data used to generate these figures are available in `/data/exports/tables/python-package-summary.csv`

Results

Across the complete dataset, Pandas is by far the most frequently requested Python package, appearing in approximately 32% of all analysed job postings. NumPy represents the second most common library (approximately 16%), followed by Matplotlib (9%) and scikit-learn (5%). All remaining packages occur only occasionally, each appearing in fewer than 3% of the analysed jobs.

Although Pandas dominates every analysed category, noticeable differences emerge between specialisations. Machine Learning jobs exhibit the broadest collection

of requested libraries, with frequent mentions of NumPy, scikit-learn, Matplotlib, OpenCV, XGBoost and LightGBM. Data Analyst jobs show a similar but slightly more analysis-oriented profile, where Pandas and NumPy dominate, accompanied by visualisation libraries such as Matplotlib, Seaborn and Plotly.

Data Engineering jobs rely overwhelmingly on Pandas, while the remaining libraries appear only sporadically. Python Automation jobs display the narrowest technology profile, where Pandas is accompanied primarily by browser automation libraries such as Selenium and Playwright.

Discussion

The results indicate that Pandas constitutes the fundamental Python library across all analysed freelance domains, regardless of job specialisation. NumPy also appears consistently, particularly in jobs involving data analysis and machine learning, confirming its role as one of the core scientific computing libraries within the Python ecosystem.

The greater diversity of requested libraries in Machine Learning jobs reflects the wider range of technical tasks typically associated with this domain, including data processing, model development and computer vision. In contrast, Data Engineering and Python Automation jobs tend to require a smaller and more specialised collection of libraries. These observations provide an initial indication of which Python packages should be prioritised when preparing for different segments of the freelance market.

3.3 Research Question 3

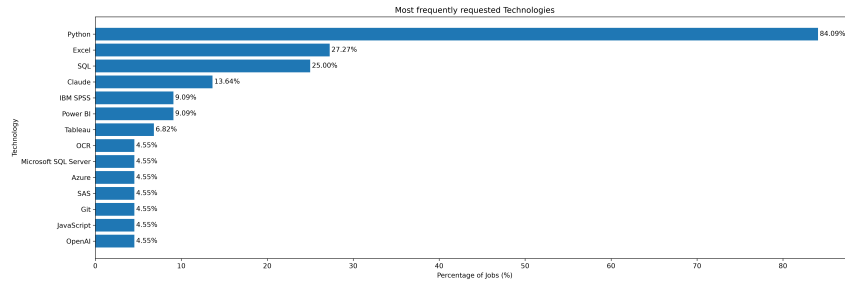
Question

Which technologies and software tools are most frequently required?

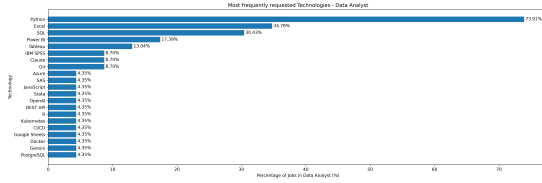
Methodology

The extraction of technologies and software tools followed the same rule-based approach used for Python package identification in the previous research question. A predefined keyword dictionary was created containing programming languages, database systems, business intelligence software, cloud platforms, AI tools, web technologies and other commonly requested technical skills. Each collected job description was automatically scanned, and every detected technology was recorded.

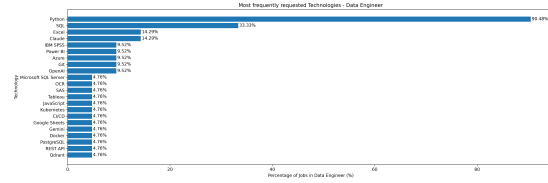
The extracted technologies were subsequently aggregated both across the entire dataset and within each job category. Their relative frequencies were calculated as the percentage of job postings mentioning each technology. The overall technology distribution and the category-specific distributions are presented in Figure 3.3, while the similarities and differences between job categories are further illustrated using the heat map shown in Figure 3.4.



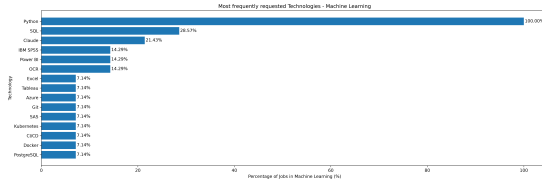
(a) Overall frequency of technologies and tools.



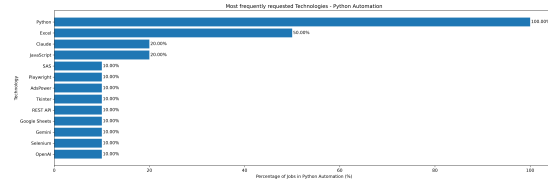
(b) Data Analyst



(c) Data Engineer



(d) Machine Learning



(e) Python Automation

Figure 3.3: Frequency of technologies and tools within the analysed dataset, overall and by job category. The complete numerical data used to generate these figures are available in `/data/exports/tables/technologies – summary.csv`

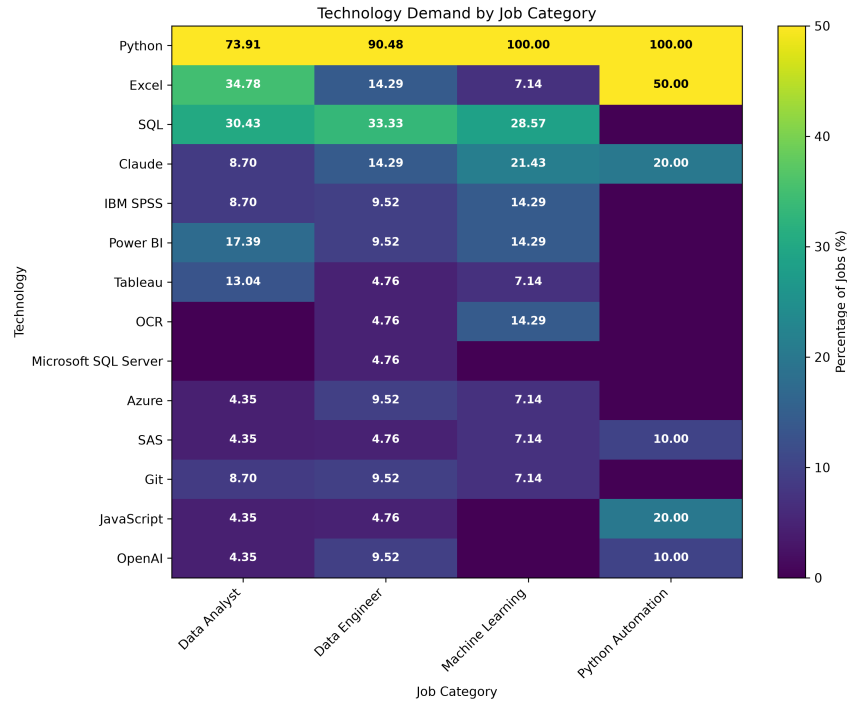


Figure 3.4: Heatmap of technology and tool usage across different job categories.

Results

The overall distribution of requested technologies is presented in Figure 3.3a. Python was by far the most frequently requested technology, appearing in more than 84% of all analysed job postings. SQL and Microsoft Excel formed the second tier of commonly required technologies, followed by business intelligence software, AI tools and cloud services.

Figures 3.3b–3.3e reveal clear differences between the analysed freelance categories. Data Analyst positions placed strong emphasis on Python together with Excel, SQL, Power BI and Tableau, reflecting their focus on data analysis and reporting tasks. Data Engineering jobs showed the highest demand for SQL and infrastructure-related technologies in addition to Python, whereas Machine Learning jobs exhibited increased usage of AI-related tools such as Claude together with OCR software and statistical packages. Python Automation jobs displayed a different profile, combining Python with spreadsheet processing, browser automation technologies and JavaScript.

The heat map shown in Figure 3.4 provides a compact comparison of these category-specific technology demands. It highlights Python as the common foundation across all analysed job categories while also revealing the complementary technologies that distinguish each freelance specialisation.

Discussion

The technology analysis demonstrates that successful freelance Python projects rarely depend on Python alone. Instead, clients typically expect developers to com-

bine Python programming with additional technologies that are closely related to the application domain.

The observed patterns suggest that Data Analyst roles require strong knowledge of analytical software and reporting tools, Data Engineering jobs demand database management and data infrastructure skills, while Machine Learning positions increasingly integrate AI platforms and specialised data processing tools. Python Automation jobs, although smaller in number, frequently combine Python with browser automation frameworks, APIs and spreadsheet-based workflows.

From the perspective of portfolio development, these findings provide practical guidance for prioritising technical skills. Rather than attempting to learn a large number of unrelated technologies simultaneously, aspiring freelancers can focus on acquiring the complementary tools that most frequently accompany Python within their chosen freelance category. Such a targeted learning strategy is likely to increase both portfolio relevance and competitiveness in the online freelance marketplace.

3.4 Research Question 4

Question

Which combinations of technologies are most frequently requested together?

Methodology

To investigate the relationships between requested technologies, every job posting was represented as a set of detected technologies obtained during the previous analysis. Pairwise co-occurrences were then calculated by counting how frequently two technologies appeared together within the same job description. The resulting frequencies were normalised by the total number of analysed jobs, and the most frequently occurring technology pairs are presented in Figure 3.5.

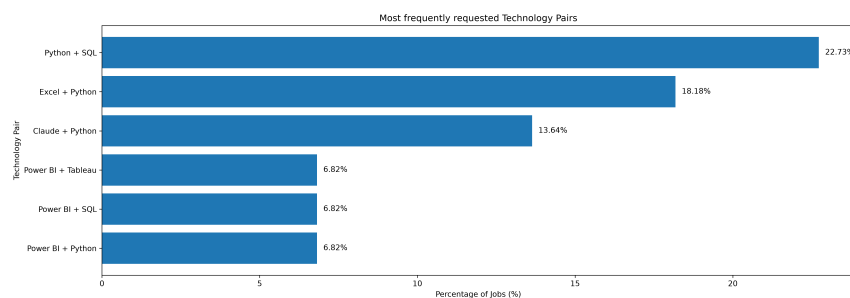


Figure 3.5: Most frequently requested technology pairs identified in the analysed Upwork job postings. The complete numerical data used to generate this figure are available in `/data/exports/tables/technology – pairs – summary.csv`

Results

The most frequently requested technology combinations are presented in Figure 3.5. The pair Python + SQL appeared most often, being mentioned in almost one quarter of all analysed job postings. The second most common combination was Python + Excel, reflecting the strong presence of data analysis and reporting tasks within the collected dataset.

Another notable result is the relatively high occurrence of the Python + Claude combination, indicating that AI-assisted workflows have already become common in Python-related freelance jobs. The remaining technology pairs, including combinations involving Power BI, Tableau and SQL, occurred considerably less frequently.

Discussion

The co-occurrence analysis demonstrates that employers typically request technology stacks rather than isolated technical skills. Python serves as the central programming language, but it is most commonly expected together with SQL, spreadsheet software or specialised domain-specific tools.

These findings reinforce the conclusions drawn from the previous research questions. Rather than focusing exclusively on Python itself, aspiring freelancers should develop complementary technical skills that frequently appear alongside Python in real client jobs. Building a portfolio around such commonly requested technology combinations is likely to better reflect actual market expectations.

3.5 Research Question 5

Question

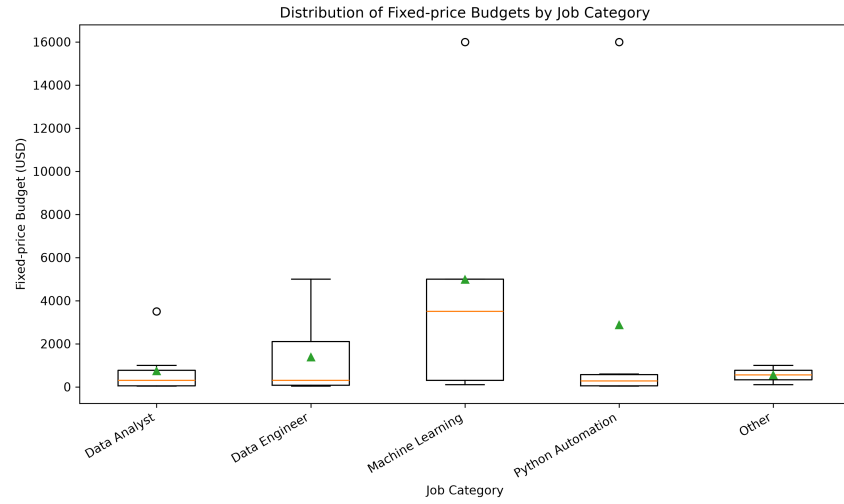
What budget ranges and hourly rates are associated with the analysed freelance jobs?

Methodology

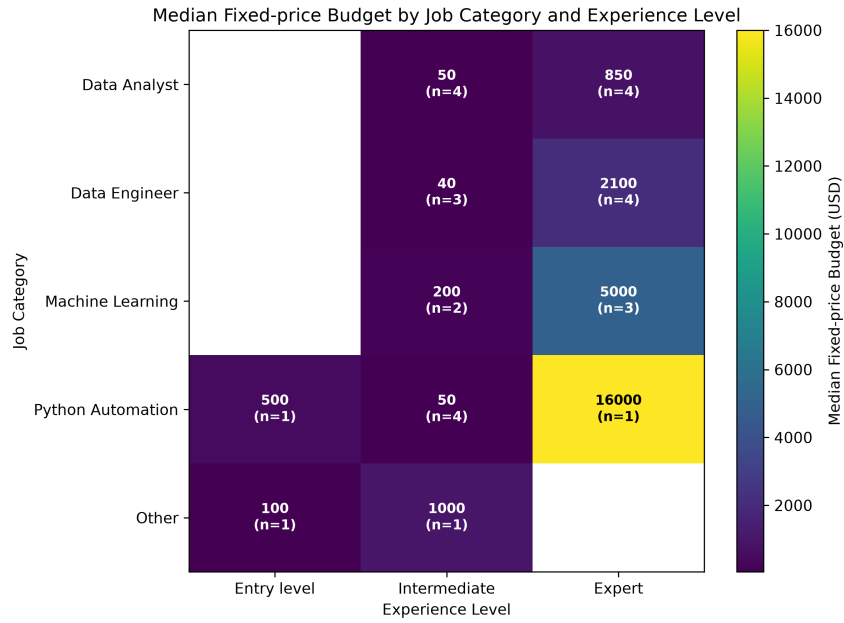
The pricing information contained in the collected Upwork job postings was analysed separately for fixed-price and hourly projects. For fixed-price contracts, the reported job budgets were grouped according to the job categories identified in the previous analyses. Likewise, hourly contracts were grouped according to their advertised hourly rates.

To better understand how compensation changes with required expertise, jobs were additionally grouped by both job category and client-requested experience level (Entry level, Intermediate and Expert). Boxplots were used to visualise the overall distribution of fixed-price budgets and hourly rates, while heatmaps summarise the median compensation for each job category and experience level. Median values were preferred over arithmetic means because they are less sensitive to extreme outliers, which are common in freelance marketplace data.

The distributions of fixed-price budgets are presented in Figures 3.6a and 3.6b, while hourly-rate statistics are shown in Figures 3.7a and 3.7b.



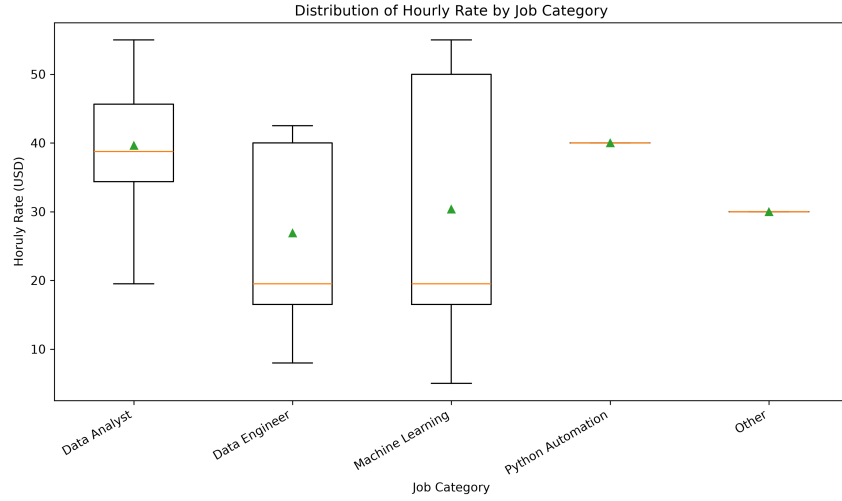
(a) Distribution of fixed-price budgets by job category.



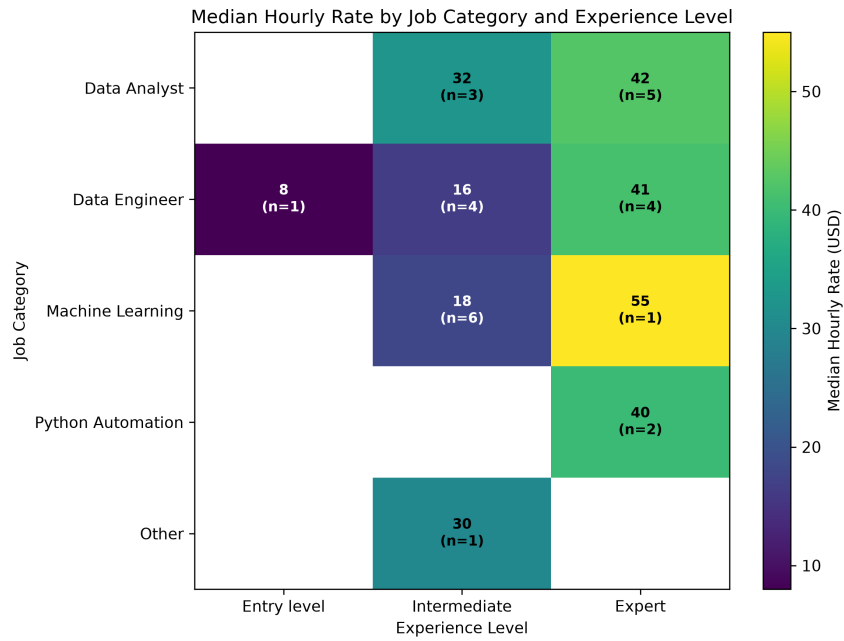
(b) Median fixed-price budgets by job category and experience level.

Figure 3.6: Budget and hourly rate distributions within the analysed dataset, overall and by job category and experience level. The complete numerical data used to generate these figures are available in */data/exports/tables/fixed – budget – summary.csv*

Results



(a) Distribution of hourly rates by job category.



(b) Median hourly rates by job category and experience level.

Figure 3.7: Budget and hourly rate distributions within the analysed dataset, overall and by job category and experience level. The complete numerical data used to generate these figures are available in */data/exports/tables/hourly – budget – summary.csv*

Results

The fixed-price budget distributions shown in Figures 3.6a and 3.6b reveal substantial differences between job categories. Machine Learning jobs generally command the highest budgets, with expert-level jobs having a median fixed price of \$5000. Data Engineer jobs also exhibit relatively high budgets, reaching a median of \$2100 for expert-level work, whereas expert-level Data Analyst jobs have a considerably lower median budget of \$850.

The hourly-rate analysis presented in Figures 3.7a and 3.7b shows a similar trend. Expert-level jobs are consistently associated with higher hourly rates than intermediate or entry-level work. The highest observed median hourly rate is \$55 for expert Machine Learning jobs, followed by approximately \$42 for expert Data Analyst jobs and \$41 for expert Data Engineer jobs. Python Automation jobs advertise hourly rates around \$40 for expert-level work.

Across both pricing models, increasing experience requirements are associated with substantially higher client budgets, although several categories contain only a limited number of observations and should therefore be interpreted with appropriate caution.

Discussion

The pricing analysis indicates that both job complexity and required expertise strongly influence freelance compensation. Machine Learning jobs offer the highest potential earnings, particularly for experienced freelancers, while Data Engineering jobs also command relatively high budgets. Data Analysis jobs appear more numerous but generally offer lower fixed-price compensation.

The results further demonstrate that experience level has a considerable effect on pricing regardless of job category. Expert-level jobs consistently receive substantially larger budgets and higher hourly rates than intermediate or entry-level work.

However, the pricing heatmaps should be interpreted with caution. The analysis is based on a dataset containing only 44 job postings, and after splitting the data simultaneously by job category and experience level, several cells contain only one or two observations. Consequently, individual median values may be strongly influenced by a single job and should therefore not be interpreted as representative market prices.

Nevertheless, despite the limited sample size, the overall trends remain consistent across both the fixed-price and hourly-rate analyses. The results suggest that jobs requiring higher levels of expertise generally command substantially higher compensation, while Machine Learning and Data Engineering appear to represent the highest-value specialisations within the analysed dataset.

3.6 Phase Summary

The analyses performed in this phase provide a comprehensive description of the analysed Python freelance market. Together, the five research questions examined market demand from complementary perspectives, including job category distribution, required Python packages, broader technology stacks, commonly requested technology combinations, and project pricing. Rather than drawing practical recommendations, the objective of this phase was to establish a reliable empirical foundation based on real Upwork job postings. These findings form the basis for the next phase, where the collected evidence will be synthesised into actionable market insights.

4 Phase 3 - Market Insights

4.1 Objectives

Phase 2 focused on collecting and analysing empirical data from Python-related freelance job postings. While the previous phase answered the individual research questions separately, the objective of Phase 3 is to transform those findings into a coherent set of market insights that can support decision making.

Rather than introducing new analyses, this phase synthesises the results of the previous chapter to identify recurring market patterns, determine the most valuable technical skills, compare different Python career paths, and highlight the technologies that appear most relevant for entry-level freelancers.

The resulting market insights provide the foundation for Phase 4, where they will be used to define the requirements of a market-oriented portfolio project roadmap.

4.2 Synthesis of the Market Analysis

The previous phase investigated the analysed freelance market from five complementary perspectives. Each research question focused on a different aspect of the collected job postings, ranging from the overall distribution of Python-related job categories to technology requirements and job compensation.

Individually, these analyses provide valuable information about specific market characteristics. However, considered together, they reveal a coherent picture of the technical skills, technologies and market trends that are most relevant for Python freelancers. Table 4.1 summarises the principal findings of each research question together with their practical interpretation.

Table 4.1: Summary of the principal findings obtained during the market analysis.

RQ	Main Finding	Implication
RQ1	Data Analysis is the largest job category.	Entry-level opportunities are concentrated primarily in Data Analysis.
RQ2	Pandas and NumPy dominate package requirements.	These libraries represent essential Python skills across multiple domains.
RQ3	Python, SQL and Excel appear most frequently.	A competitive freelancer should master the core technology stack rather than Python alone.
RQ4	Python is commonly requested together with SQL and Excel.	Portfolio projects should demonstrate the integration of complementary technologies.
RQ5	Machine Learning and Data Engineering generally offer higher compensation.	Specialisation may provide greater long-term earning potential.

The synthesis presented in Table 4.1 demonstrates that the five research questions complement one another rather than providing independent observations. The analyses collectively indicate that successful Python freelancers require a combination of widely applicable programming skills and domain-specific expertise.

The findings also reveal that market demand cannot be characterised by a single technology or programming library. Instead, employers consistently expect practical combinations of technologies, particularly Python together with SQL, spreadsheet software, and widely adopted data-analysis libraries. Furthermore, although Data Analysis represents the largest proportion of the analysed market, the pricing analysis suggests that more specialised fields, such as Machine Learning and Data Engineering, may offer greater long-term earning potential.

These observations provide the basis for identifying the core technical skills required by the market, which are examined in the following section.

4.3 Core Skills Required by the Market

The previous sections identified the technologies, programming languages, software tools, and libraries most frequently requested in the analysed Upwork job postings. Rather than presenting additional statistics, this section synthesizes these findings into a practical hierarchy of technical skills. The objective is to distinguish between universally required competencies and technologies that are valuable only for specific job types. This hierarchy provides the foundation for the portfolio project design presented in the next phase.

Table 4.2: Skill hierarchy derived from the market analysis.

Skill Level	Technologies
Essential	Python, Pandas, NumPy, SQL, Excel
Important	Power BI, Tableau, Git, Docker, Azure, REST API
Specialized	Machine Learning libraries (Scikit-learn, XGBoost, LightGBM, OpenCV), Selenium, Playwright, OCR frameworks, IBM SPSS, SAS, JavaScript, PostgreSQL, Microsoft SQL Server, Kubernetes, CI/CD, OpenAI APIs, Claude APIs, Gemini APIs and other domain-specific technologies.

The market analysis suggests that a relatively small set of technologies forms the foundation of most freelance data-related jobs. Python clearly represents the central programming language, while Pandas and NumPy constitute the most consistently requested Python libraries. SQL and Excel also appear across nearly every analysed job category, indicating that database querying and spreadsheet-based data manipulation remain essential practical skills.

A second tier of technologies consists of tools that substantially increase employability but are not required for every job. These include business intelligence platforms such as Power BI and Tableau, collaborative development tools such as Git, deployment technologies such as Docker, cloud platforms including Azure, and REST APIs for interacting with external services.

Finally, a number of specialised technologies appear only within particular domains. Machine Learning jobs frequently require dedicated frameworks such as Scikit-learn, XGBoost or LightGBM, while automation jobs introduce tools such as Selenium and Playwright. Similarly, cloud infrastructure, OCR libraries, large language model APIs, and orchestration technologies occur only in subsets of the analysed jobs.

This hierarchy indicates that portfolio development should prioritise mastery of the essential technologies before expanding toward specialised domains. A strong foundation built on Python, SQL, Pandas, NumPy and Excel is therefore expected to maximise applicability across the largest proportion of freelance opportunities.

4.4 Comparison of Python Career Paths

The previous sections identified the most common job categories, technologies, required skills, and compensation patterns within the analysed Upwork dataset. This section combines these findings to provide a qualitative comparison of the four primary Python-related career paths represented in the collected job postings. Rather than focusing on individual statistics, the objective is to characterise each market segment from the perspective of a prospective freelance developer.

Table 4.3: Comparison of the major Python-related freelance career paths.

Category	Demand	Complexity	Compensation	Typical Technologies
Data Analysis	High	Medium	Medium	Python, Pandas, NumPy, SQL, Excel, Power BI
Data Engineering	Medium	High	High	Python, SQL, Docker, Azure, REST APIs
Machine Learning	Medium	Very High	High	Python, Scikit-learn, XGBoost, LightGBM, OpenCV
Python Automation	Low	Medium	Medium	Python, Selenium, Playwright, JavaScript

The market analysis reveals that the analysed Python freelance market is composed of several distinct career paths with different characteristics.

Data Analysis represents the largest and most accessible segment of the market. It contains the highest number of job postings and relies primarily on a relatively stable collection of widely adopted tools, including Python, Pandas, NumPy, SQL, and Excel. Although compensation is generally moderate compared with more specialised domains, the high demand makes this category particularly attractive for beginners seeking to build practical experience.

Data Engineering occupies a smaller but more technically demanding segment. jobs frequently involve database management, cloud platforms, APIs, and deployment technologies. While the required technical complexity is noticeably higher, the analysed jobs also indicate a tendency toward larger budgets and higher hourly rates.

Machine Learning forms the most specialised category within the analysed dataset. In addition to the common Python ecosystem, these jobs require dedicated machine learning libraries and domain-specific knowledge. Although the number of available jobs is lower than for Data Analysis, the observed budgets suggest that advanced expertise is generally rewarded with higher compensation.

Python Automation appears as the smallest of the four primary categories. Most jobs focus on browser automation, scripting, or workflow automation using tools such as Selenium and Playwright. While the technical barrier to entry is lower than for Machine Learning or Data Engineering, the overall number of opportunities identified in the analysed sample is also considerably smaller.

Overall, the comparison suggests that the different career paths represent different trade-offs between market size, technical complexity, and earning potential. Data Analysis offers the broadest entry point into the freelance market, whereas Data

Engineering and Machine Learning provide opportunities for higher long-term compensation at the cost of greater technical specialisation. Python Automation occupies a niche position, serving clients with specific automation requirements rather than general data-related tasks.

Table 4.4: Key market insights derived from the analysis.

Market Insight	Practical Meaning
Python is the universal programming language	Every portfolio project should be implemented primarily in Python.
SQL is a core complementary skill	Database querying is expected across nearly every data-related role.
Python alone is rarely sufficient	Clients typically require combinations of programming, databases and business tools.
Specialisation increases earning potential	Data Engineering and Machine Learning projects generally offer higher compensation.
Technology integration is highly valued	Successful projects combine multiple complementary technologies rather than demonstrating isolated skills.

The findings summarised in Table 4.4 represent the principal conclusions of the market analysis. Rather than describing individual technologies or job categories, these insights identify the characteristics that consistently distinguish successful freelance projects. Consequently, these observations provide the design principles for the portfolio roadmap developed in the following phase.

4.5 Implications for Portfolio Development

The analyses presented in the previous sections not only describe the current Python freelance market but also provide clear guidance for portfolio design. Each freelance category is characterised by a distinct combination of technologies, technical complexity and client expectations. Consequently, portfolio projects should be designed to reflect the requirements of their target market rather than demonstrating isolated programming techniques.

Data Analysis

A competitive Data Analysis portfolio should demonstrate the ability to analyse real-world datasets and communicate meaningful insights.

- use Python as the primary programming language;
- demonstrate practical use of Pandas and NumPy for data manipulation;
- integrate SQL for data extraction and querying;

- include data cleaning and preprocessing;
- perform exploratory data analysis;
- produce informative visualisations using libraries such as Matplotlib or Seaborn;
- present interpretable conclusions and recommendations based on the analysis;
- optionally integrate business intelligence tools such as Power BI or Tableau for interactive dashboards.

Data Engineering

A competitive Data Engineering portfolio should demonstrate the ability to build reliable and scalable data-processing workflows.

- integrate Python with SQL database;
- demonstrate automated ETL or ELT pipeline;
- include data validation and transformation;
- interact with external APIs;
- demonstrate version control using Git;
- include containerisation with Docker where appropriate;
- show reproducible and modular project organisation.

Machine Learning

A competitive Machine Learning portfolio should demonstrate the complete lifecycle of a predictive modelling project.

- perform data preprocessing and feature engineering;
- implement machine learning models using Scikit-learn or similar libraries;
- compare multiple modelling approaches;
- evaluate model performance using appropriate metrics;
- present clear interpretations of the model results;
- demonstrate reproducible experimentation and well-structured code.

Python Automation

A competitive Python Automation portfolio should demonstrate the ability to automate realistic business processes and repetitive tasks.

- automate data collection or processing workflows using Python scripts;
- interact with web pages or external APIs;
- demonstrate browser automation where appropriate;
- integrate spreadsheet processing where relevant;
- emphasise robustness through logging and error handling;

- produce reusable and well-documented automation scripts.

4.6 Phase Summary

Phase 3 transformed the quantitative results obtained during the exploratory market analysis into a coherent set of practical market insights. Rather than examining individual statistics in isolation, the analyses were synthesised to identify the core technical skills required by the market, compare the major Python-related freelance career paths, and establish category-specific requirements for competitive portfolio projects.

Although the four freelance categories differ considerably in their technical focus, several common characteristics emerged from the market analysis. Successful portfolio projects should demonstrate practical problem solving, clean software engineering practices, realistic datasets, and the integration of multiple complementary technologies. At the same time, each career path requires its own combination of domain-specific tools and workflows, which should be reflected in the portfolio.

Consequently, the market insights developed throughout this phase establish a set of design requirements rather than specific project ideas. These requirements provide the foundation for the next phase, in which a portfolio roadmap will be constructed by selecting projects that satisfy the identified market needs while covering the most relevant technologies and skills.

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